Mechanistic Estimation of Random ReLU MLPs: Deep Third-Cumulant Representations under Hard FLOP Budget

Research notes in response to the factorized K=3 chain problem (n=1024, L=16, B=2^{41} FLOPs). Claims labelled D = derived, C = conjecture, M = to be measured.

Q1. Representations for deep (old) third-cumulant content

All proposals stay inside the factorized-source / Hermite-closure setting already used by V29. Costs use the stated unit (one dense n×n×n matmul = 2^{31} FLOPs, n=1024). "Young" work (ages ≤4) is left untouched; only the old tier is replaced.

(a) Hermite expansion of the ReLU map inside the shared low-rank basis Q_c (moves nonlinearity into coordinates)

Object carried for each old source s: the three thin matrices F_A, $F_P \in \mathbb{R}^{r\times n}$ (r ≈ 150–220) already used by V29, plus the r×r Gram $G = Q_c^T C Q_c$ and the diagonal first/second Hermite coefficients w_1, w_2 expressed in the same basis.

Update (exact transport of legs + approximate closure):

```
F_P ← (Q_c^T W diag(w_1) Q_c) F_P, F_A ← F_P a_b
(the matrix in parentheses is r×r). The needed slices are then formed by a degree-3 Hermite
polynomial in the projected variables:
D_21 ≈ Q_c (contraction of F_A, F_P with w_2 and the 3-tensor of G) Q_c^T
(the only n-scale work is the final multiplies by Q_c).
```

Cost per layer for the whole old tier (S ≈ 10 sources, r = 192): ≈ 2.1 units (the r^3 and $n\,r^2$ terms dominate; Q_c itself is amortized). Relative to exact transport the method loses the high-rank residual of the Hadamard products that live outside range(Q_c). By the measured spectra that residual carries >80% of the Frobenius energy of A∘P, so the approximation is controlled by how much of the signal (not the energy) lives in the top r directions of the products.

First falsifier (one network, <10 min CPU): recompute exact D_{21} contributions of all sources older than age 5 at layers 8,10,12,14; project the same contributions onto the current Q_c and measure relative RMS error. If it exceeds 3% at any layer the method is dead (the 2% budget required by the empirical error law is already violated).

(c) Exact transport of the needed slices themselves, closed by a low-rank remainder for the off-slice part of κ_3

Object: the pair ($D_{21} \in \mathbb{R}^{n\times n}$, $D_3 \in \mathbb{R}^n$) for the aggregate of all sources older than age 4, together with the current covariance C. (No per-source legs after age 4.)

Exact transport identity for a single source (D, from the chain rule + Hermite):

```
D_21^(l+1) = (W diag(w_1)) D_21^(l) (W diag(w_1))^T
+ terms involving only D_3 and the diagonal of C
+ R^(l) (remainder from j ≠ k indices).
```

The remainder R is itself a contraction of the full κ_3 against two legs of W. For old sources the parity identity of §3b implies that the conditional expectation of every odd object vanishes, so R can be replaced by its projection onto the even algebra generated by C:

```
R ≈ λ_1(C ∘ C) + λ_2 diag(C) C + λ_3 diag(C)^2
(three scalar coefficients per layer, fixed once by a short Monte-Carlo on a single network).
Aggregate over the old tier and the same three scalars are used for all of them.
```

Cost per layer: one dense covariance transport (≈0.7 units) + two dense n×n multiplies for the D_{21} transport (≈1.4 units) + $O(n^2)$ element-wise work = 2.2 units. Loss relative to exact transport: only the part of R that is not spanned by the three even monomials of C.

First falsifier (<30 min): at layers 8–14 of one network compute the exact residual R of every old source, project it onto the three monomials, and measure the relative Frobenius error of the projected D_{21}. If any layer exceeds 2% the closure is insufficient.

(d) Identity that evaluates $(A \circ P)A^T$ at cost $O(n\, r^2)$ without materialising dense legs

Let $A = Q_c\, F_A$, $P = Q_c\, F_P$. The Hadamard product cannot be formed in factor space, but the bilinear form that appears in every needed contraction can:

```
((A ∘ P) A^T) v = Q_c ( F_A ( (F_P^T Q_c^T) ∘ (F_A^T Q_c^T v) ) )
(the Hadamard is now only r×n). All four contractions that produce D_21 and D_3 admit analogous
rewrites (D, pure linear algebra).
```

Cost for the whole old tier: $4\, S \cdot n\, r^2 \approx 1.8$ units at r=192. The only approximation is the rank truncation of the legs themselves (already known to be safe down to the measured rank law).

First falsifier: implement the rewritten contractions on the stored dense legs of layers 10 and 14 of one network and verify bit-for-bit agreement with the original V29 code (must match to machine precision; any discrepancy kills the identity).

No other representation that simultaneously respects parity, the measured singular spectra, and the per-layer sensitivity map of §3b is visible at present.

Q2. Three ranked hypotheses for what the leaders carry instead of the old-source tier

1. Aggregate-slice transport with a learned or analytic even closure for the remainder (exactly proposal (c) above).

Cost signature: total bill ≈ young tier (140–160 units) + 2–3 units per layer for the old aggregate. Accuracy signature: raw MSE in the $1.6–1.9\times10^{-8}$ band. Distinguishing local experiment: replace V29's old tier by the three-scalar closure and measure the final MSE; if it lands inside the leaders' band the hypothesis is confirmed.

2. Single shared propagator for the whole old tier.

All old sources are transported by one common dense matrix P_{old} (or a low-rank factorisation of it) multiplied by per-source scalar amplitudes that evolve by the diagonal w_1. The needed slices are then rank-1 updates. Cost ≈ 0.8 units/layer. The hypothesis predicts that the relative amplitudes stay almost constant after age 6 (because of the parity-driven mean-zero property). Falsifier: compute the principal angles between the column spaces of the dense legs of different old sources at layer 12; if the angles are not $O(1/\sqrt{n})$ the shared-propagator assumption fails.

3. Quenched orthogonal-polynomial expansion of the whole layer map that never separates sources.

The state is the pair $(C, \{\mu_k\}_{k=1}^{3})$ where μ_k are the first three Hermite coefficient tensors of the push-forward measure, kept only through their action on the needed slices. Cost of the cubic term is absorbed into the same n×n multiplies already paid for covariance transport, giving a total bill indistinguishable from the young tier alone. Distinguishing observable on the leaderboard: the method should degrade gracefully when the network depth is increased beyond 16 while keeping width fixed (the polynomial degree stays 3); pure source-tracking methods degrade linearly with the number of live sources.

Q3. Most informative next measurement

Proposals (a), (c), (d) are all credible on the evidence given; none can be dismissed a priori. The single most informative next measurement is therefore the residual spectrum of proposal (c): for every old source at layers 8, 10, 12, 14 compute the exact remainder R, its projection onto the three even monomials of C, and the resulting relative error in D_{21}. If that error stays ≤ 2% the aggregate-slice route is viable at ≈ 2 units/layer and immediately explains the leaders' cost; if it is larger the same data will show which additional even tensors are required, giving a concrete target for the next closure.

Cost unit reminder: 1 unit = one dense (1024×1024×1024) matmul = 2^{31} FLOPs. Budget B = 1024 units. V29 baseline = 260 units; leaders = 133–165 units. Young tier alone ≈ 153 units.